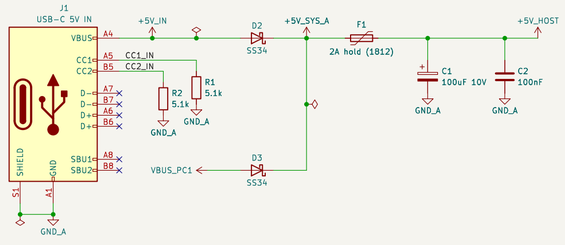
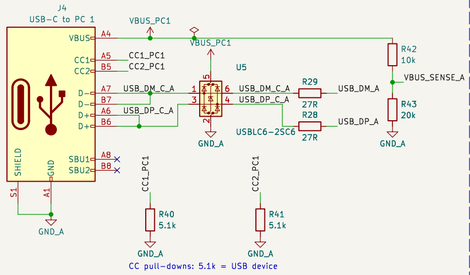
Power input: adapter (J1) OR-ed with PC 1 VBUS

D2 / D3 let the adapter and PC 1 share the load without back-feeding each other. F1 protects the USB-A port.



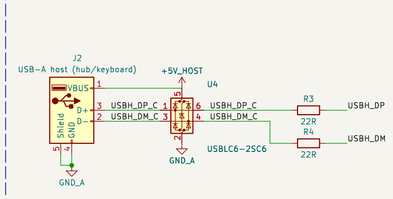
PC 1: USB-C device port + VBUS sense

MCU-A enumerates on PC 1 (27 ohm series R28 / R29, ESD array U5).
GPIO24 sees PC 1 VBUS through R42 / R43.



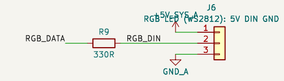
USB host port: keyboard, mouse, dongle or hub

USBH_DP / USBH_DM come from MCU-A GPIO2 / GPIO3 (PIO-USB host).
U4 (USBLC6-2SC6) clamps ESD on the port.



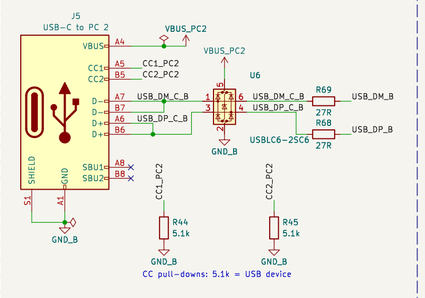
Optional RGB LED header J6

WS2812 / SK6812: 5V, DIN, GND. Data from GPIO6 through R9.

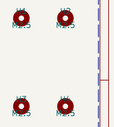


PC 2: USB-C device port (powers side B)

MCU-B enumerates on PC 2 (27 ohm series R68 / R69, ESD array U6) and runs from its VBUS.

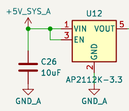


Mounting holes

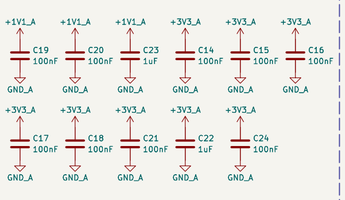


MCU-A: RP2040 (USB host, router, microSD; enumerates on PC 1)

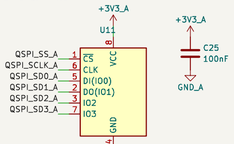
3.3 V LDO

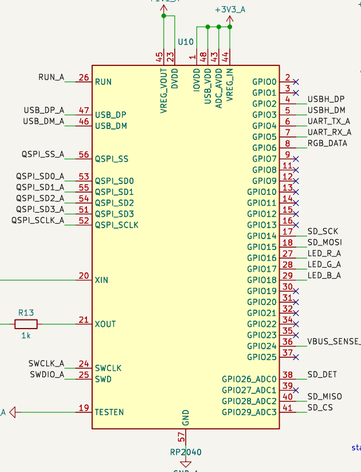


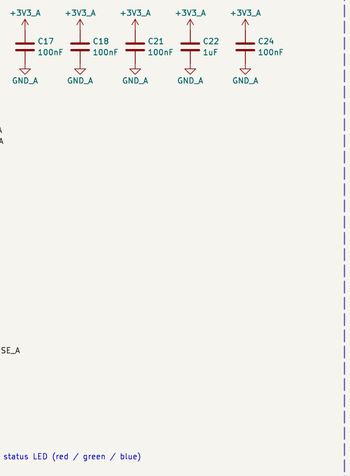
decoupling, one per supply pin:
DVDD x2, VREG_VOUT, IOVDD x6, USB_VDD, VREG_IN, ADC_AVDD



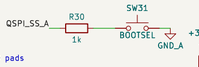
2 MB QSPI flash





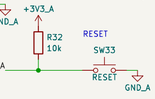


BOOTSEL

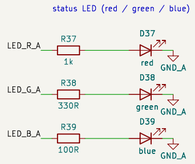


SWD pads



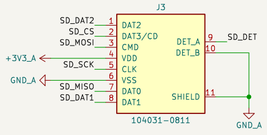


status LED (red / green / blue)



microSD: shared drive (SPI1)

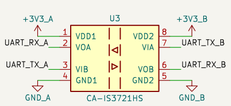
Shown as a USB drive on the active PC only.
10k pull-ups on CS, MISO and the unused DAT1 / DAT2.





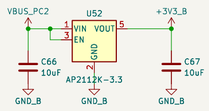
Isolation: GND_A and GND_B never meet

CA-IS3721HS (pin compatible with ADuM1201).

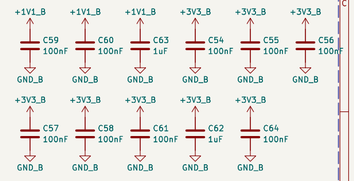


MCU-B: RP2040 (enumerates on PC 2, powered by PC 2)

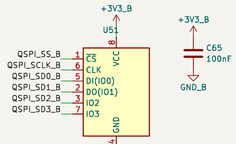
3.3 V LDO

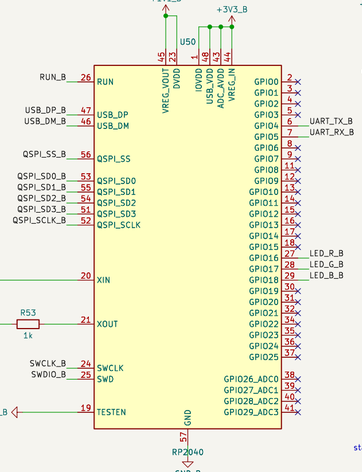


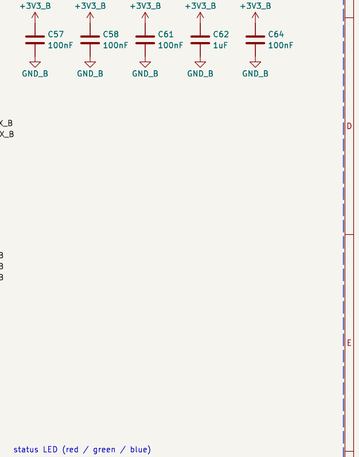
decoupling, one per supply pin:
DVDD x2, VREG_VOUT, IOVDD x6, USB_VDD, VREG_IN, ADC_AVDD



2 MB QSPI flash





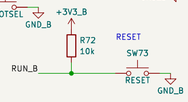


BOOTSEL

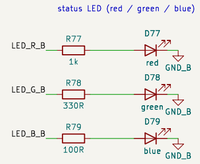


SWD pads





status LED (red / green / blue)



Notes

GPIO map (same as variant A, one firmware for both)
  GPIO2 / GPIO3: USB host D+ / D- (PIO-USB, 22 ohm series). GPIO4 / GPIO5: UART1 TX / RX through the isolator.
  GPIO14 / GPIO15 / GPIO28 / GPIO29: SPI1 SCK / MOSI / MISO and card CS. GPIO26: card detect. GPIO6: external RGB LED (J6).
  GPIO24: PC 1 VBUS sense (10k / 20k divider). GPIO16 / 17 / 18: status LED red / green / blue.

Power domains
  side A (GND_A): adapter (J1) and PC 1 VBUS (J4) are OR-ed by D2 / D3 into +5V_SYS_A, which feeds the USB host port (through F1),
    the RGB header and LDO U12 (+3V3_A, MCU-A, microSD). Neither source can back-feed the other.
  side B (GND_B): PC 2 VBUS (J5) feeds LDO U52 (+3V3_B, MCU-B). The only link between the sides is the isolator U3.

First flash: hold BOOTSEL, tap RESET, copy the .uf2 to the drive that appears. SWD pads (SWCLK, SWDIO, GND) for a debugger.

kmswitch (open hardware)	
Sheet: /	
File: kmswitch-b.kicad_sch	
Title: kmswitch — variant B (2x RP2040 chip-down)	
Size: A2	Date:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	**Rev: 1.0**
	Id: 1/1